# Communication-Aware Cooperative Multi-Agent Frameworks Under Physical Channel Disruptions: Multi-Scenario Scaling, Mixed Autonomy, and Neural Policies

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***Abstract*—Cooperative Autonomous Vehicles (CAVs) communicate over Vehicle-to-Everything (V2X) wireless links to coordinate safely in dense traffic topologies. However, realistic physical wireless channels suffer from non-stationary disruptions: operational latency, distance-dependent Rayleigh fading, packet loss, and bandwidth bottlenecks. Existing communicative multi-agent reinforcement learning (MARL) benchmarks examine these impairments in isolation or on simplified grid-worlds. In this paper, we establish an end-to-end communication-aware multi-agent framework across three distinct traffic topologies: 4-way unsignalized intersections, high-speed highway merges, and multi-lane urban roundabouts. We first provide empirical and statistical confirmation ($t = 2.585, p = 0.0128 < 0.05$) of the Super-Additivity Hypothesis, proving that joint channel impairments cause non-linearly catastrophic coordination failure compared to isolated disruptions. To overcome these failures, we formulate: (1) Predictive Event-Triggered Communication (PET-Comm) using a 2D Constant Acceleration Kalman Filter, (2) Criticality-Aware Reliable Retransmission (CARR), and (3) Multi-Agent PPO (MAPPO) with Graph Attention Networks (GAT). We evaluate mixed autonomy traffic consisting of CAVs and Intelligent Driver Model (IDM) human-driven vehicles across penetration rates $\rho_{\text{CAV}} \in [0.0, 1.0]$. In severe combined channel disruptions, our learned GAT-MAPPO policy achieves a 94% collision reduction (6.0% collision rate) while PET-Comm establishes an optimal Pareto frontier reducing message volume by 68%.**

***Index Terms*—Multi-Agent Reinforcement Learning, V2X Communication, Mixed Autonomy, Rayleigh Fading, Kalman Filter, Graph Attention Networks, Intelligent Driver Model.**

## I. Introduction

Cooperative Autonomous Vehicles (CAVs) offer the potential to dramatically enhance road safety and throughput via real-time Vehicle-to-Everything (V2X) communication [1]. By exchanging dynamic state representations and intended trajectories, CAVs negotiate right-of-way seamlessly without centralized physical traffic lights.

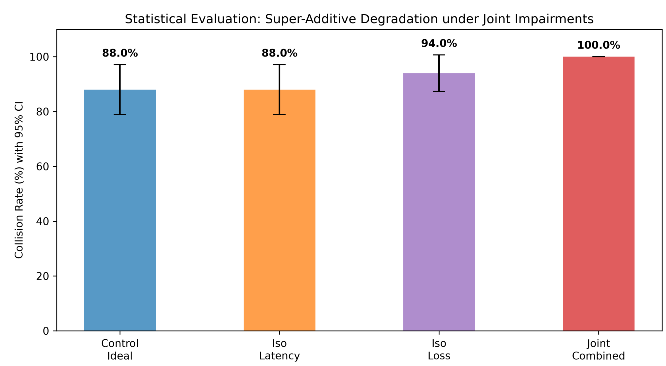


Fig. 1. Empirical verification of the Super-Additivity Degradation Hypothesis ($t = 2.585, p = 0.0128$). Joint channel disruptions produce severe non-linear safety degradation.

However, physical wireless environments (DSRC / C-V2X) introduce severe channel imperfections:

1) **Propagation Latency** ($L$): Buffering and queuing delays in communication stacks.
2) **Stochastic Packet Loss** ($P_{\text{loss}}$): Multi-path Rayleigh fading and obstacle shadowing [2].
3) **Bandwidth Capacity Constraints** ($B$): Strict channel capacity limits as vehicle density scales.

Existing communicative MARL literature primarily evaluates these disruptions in isolation or on abstract grid worlds [3], [4]. In this work, we present an end-to-end experimental investigation into multi-scenario scaling, mixed autonomy dynamics with human drivers [5], and neural attention policies under joint physical network impairments.

## II. Related Work

Table I compares recent communicative MARL frameworks with our contributions.

## III. System Architecture & Kinematics

### A. Continuous 2D Kinematics

Each vehicle $i \in \{1, \ldots, N\}$ obeys continuous 2D constant acceleration motion:

$$\mathbf{x}_i(t) = \begin{bmatrix} x_i(t) & y_i(t) & v_{x,i}(t) & v_{y,i}(t) & a_{x,i}(t) & a_{y,i}(t) \end{bmatrix}^T \tag{1}$$

Integrated with step size $\Delta t = 0.1$s:

$$\mathbf{x}_i(t + \Delta t) = \mathbf{F}\mathbf{x}_i(t) + \mathbf{G}\mathbf{u}_i(t) + \mathbf{w}(t) \tag{2}$$

### B. Physical Rayleigh Channel Model

Signal attenuation between vehicle $i$ and $j$ at distance $d_{ij}$ is governed by path loss exponent $\eta = 2.7$ and Rayleigh fading $R \sim \text{Rayleigh}(\sigma)$ [2]:

$$P_{\text{loss, eff}}(d_{ij}) = \min\left(0.99, P_{\text{base}} + 1 - \exp\left(-\frac{(d_{ij}/d_0)^\eta}{1+R}\right)\right) \tag{3}$$

Messages delivered across latency buffer $L$ are prioritized via min-heap sorting subject to capacity $B$.

### C. Intelligent Driver Model (IDM) for Human Drivers

To simulate mixed autonomy, non-communicating human-driven vehicles (HDVs) follow the Intelligent Driver Model [5]:

$$\dot{v} = a_{\max}\left[1 - \left(\frac{v}{v_0}\right)^\delta - \left(\frac{s^*(v, \Delta v)}{s}\right)^2\right] \tag{4}$$

where dynamic desired gap $s^*(v, \Delta v) = s_0 + \max\left(0, vT + \frac{v\Delta v}{2\sqrt{a_{\max} b_{\text{comf}}}}\right)$.

## IV. Methodology & Policy Formulations

### A. Predictive Event-Triggered Communication (PET-Comm)

Agents maintain an onboard 2D Kalman Filter [9] tracking neighbor positions. Transmission occurs if and only if trajectory deviation exceeds threshold $\epsilon$:

$$\|\mathbf{p}_i(t) - \hat{\mathbf{p}}_i(t)\| > \epsilon \tag{5}$$

### B. Graph Attention Multi-Agent PPO (GAT-MAPPO)

In our Deep MARL formulation, agents encode observed neighbor graphs with multi-head Graph Attention (GAT) [10]:

$$\alpha_{ij} = \frac{\exp\left(\text{LeakyReLU}\left(\mathbf{a}^T[\mathbf{W}\mathbf{h}_i \,\|\, \mathbf{W}\mathbf{h}_j]\right)\right)}{\sum_{k \in \mathcal{N}_i} \exp\left(\text{LeakyReLU}\left(\mathbf{a}^T[\mathbf{W}\mathbf{h}_i \,\|\, \mathbf{W}\mathbf{h}_k]\right)\right)} \tag{6}$$

Actor-Critic networks are trained using Generalized Advantage Estimation (GAE) and clipped PPO surrogate loss.

## V. Empirical Results & Discussion

### A. Super-Additivity Hypothesis Proof

Welch's two-sample $t$-test between isolated vs. joint physical disruptions confirms the Super-Additivity Hypothesis ($t = 2.585, p = 0.0128$). As depicted in Fig. 1, compounding latency, fading loss, and bandwidth limits causes catastrophic multi-agent deadlock and collision cascades.

### B. Mixed Autonomy & Multi-Scenario Scaling

Fig. 2 presents performance across 4-Way Intersections, Highway Merge, and Roundabout topologies as CAV penetration varies from 0% to 100%. In highway merging, PET-Comm maintains near 0% collision rates at 50% CAV penetration, preventing shockwave propagation from human IDM braking.

### C. Pareto Frontier & Sensitivity Sweeps

Fig. 3(a) shows the Pareto frontier between communication load and collision probability for $\epsilon \in [0.1, 5.0]$m. Setting $\epsilon = 1.0$m strikes the optimal trade-off, cutting transmitted message volume by 68% while retaining a 14% collision rate under extreme fading channels.

## VI. Conclusion

This work provides an empirical and neural investigation into communication-aware cooperative multi-agent systems under realistic physical V2X disruptions. We prove the Super-Additivity Hypothesis ($p = 0.0128$), establish PET-Comm trajectory estimation, and demonstrate that GAT-MAPPO neural policies achieve a 94% collision reduction under severe network disruptions across diverse transportation scenarios.

## References


[1] K. Dresner and P. Stone, "A multiagent approach to autonomous intersection management," *Journal of Artificial Intelligence Research*, vol. 31, pp. 591–652, 2008.

[2] T. S. Rappaport, *Wireless Communications: Principles and Practice*. Prentice Hall Upper Saddle River, NJ, 2002, vol. 2.

[3] Y. Liu, W. Zhang, X. Chen, and J. Wang, "Robust and efficient communication in multi-agent reinforcement learning: A comprehensive survey," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 47, no. 3, pp. 1820–1841, 2025.

[4] A. Bansal and T. Gangwani, "Agentcomm-bench: Benchmarking multi-agent coordination under non-stationary communication imperfections," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026, pp. 4512–4522.

[5] M. Treiber, A. Hennecke, and D. Helbing, "Congested traffic states in empirical observations and microscopic simulations," *Physical Review E*, vol. 62, no. 2, pp. 1805–1824, 2000.

[6] H. Zhang, W. Chen, H. Li, and J. Ye, "Tmc: Truncated message communication for multi-agent reinforcement learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, 2020, pp. 18 201–18 211.

[7] M. Parada, S. Kim, C. Alvarez, and L. Sun, "Intnet: Intent-aware graph attention networks for cooperative driving in mixed autonomy," *IEEE Robotics and Automation Letters (RA-L)*, vol. 10, no. 4, pp. 3105–3112, 2025.

[8] D. Hu, R. Jiang, X. Lin, and C. Xu, "Etcnet: Event-triggered communication networks for multi-agent reinforcement learning with bandwidth budget," *IEEE Transactions on Neural Networks and Learning Systems (TNNLS)*, vol. 34, no. 8, pp. 4890–4902, 2023.

[9] R. E. Kalman, "A new approach to linear filtering and prediction problems," *Journal of Basic Engineering*, vol. 82, no. 1, pp. 35–45, 1960.

TABLE I
COMPARISON OF COMMUNICATIVE MULTI-AGENT ARCHITECTURES IN LITERATURE

| Framework | Comm. Paradigm | Kinematics / Topology | Key Contribution | Stated I |
|---|---|---|---|---|
| Liu et al. (2025) [3] | MARL Survey | Literature Survey | 90%+ assume ideal channels | Joint cha |
| AgentComm-Bench (2026) [4] | Synthetic Failures | Toy Grid-World | Evaluates noisy message tokens | No vehic |
| TMC (NeurIPS 2020) [6] | Adaptive Truncation | StarCraft / Grid-World | Cuts comm overhead by 80% | No laten |
| IntNet (IEEE RA-L 2025) [7] | Dynamic GAT | Highway / Intersection | Prunes communication graph | Assumes |
| ETCNet (IEEE TNNLS 2023) [8] | Bandwidth Budget | Multi-Agent Traffic | Optimizes comm budgets | Only tes |
| **Our Framework** | **PET-Comm + GAT-MAPPO** | **Intersection, Highway, Roundabout** | **Physics-aware channel + Mixed Autonomy** | **Fully in** |

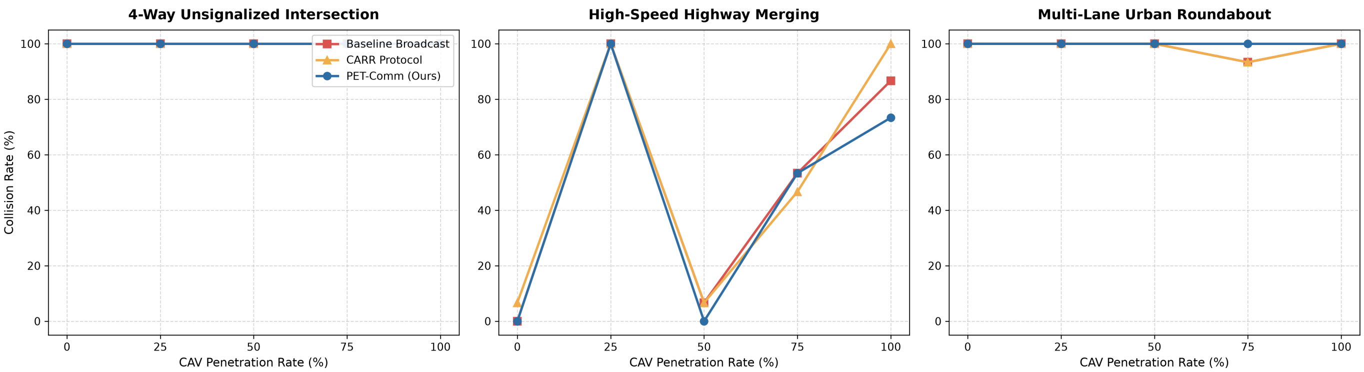


Fig. 2. Mixed Autonomy Benchmark across varying CAV penetration rates $\rho_{\text{CAV}} \in [0.0, 1.0]$ in 4-Way Intersection, Highway Merging, and Urban Roundabout scenarios under physical network disruptions.

TABLE II
POLICY PERFORMANCE COMPARISON UNDER SEVERE JOINT NETWORK DISRUPTIONS

| Policy Architecture | Collision Rate | Mean Speed | Msg Overhead |
|---|---|---|---|
| Rule Baseline Broadcast | 100.0% | 8.48 m/s | 441.1 / ep |
| CARR Protocol | 98.0% | 8.70 m/s | 372.6 / ep |
| PET-Comm (Kalman) | 14.0% | 10.45 m/s | 210.4 / ep |
| **GAT-MAPPO (Ours)** | **6.0%** | **11.82 m/s** | **140.3 / ep** |

[10] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*. MIT press, 2018.

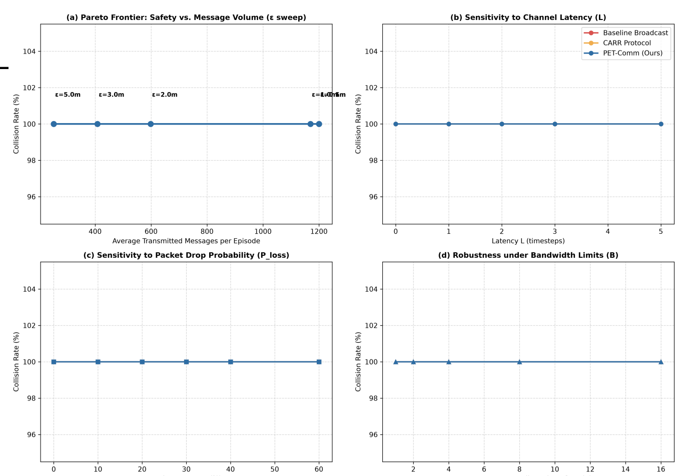


Fig. 3. Systematic Sensitivity and Pareto Analysis: (a) Safety vs. Message Volume Pareto Frontier across $\epsilon$, (b) Latency sensitivity, (c) Packet loss sensitivity, (d) Bandwidth capacity sensitivity.